Compositional probabilistic scoring and tie-band-aware benchmarking for methylome-guided ThermoCas9 target-site ranking

Technical memo from the thermocas methylome-analysis framework (audit-trail revision)

Allison Huang (Columbia University)

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Abstract

Methylation-sensitive Cas9 target selection is a ranking problem: candidates should be hypomethylated in disease cells, methylated in the matched normal comparator, and supported by trustworthy methylation evidence. We present a compositional probability-scale scoring skeleton, `p_targ × (gap factor) × p_trust`, paired with a tie-band-aware benchmark contract (`precision_at_k_{min,max}`, `tie_band_size_at_k`, mid-rank Mann-Whitney AUC). Two gap factors ship in this tag: V2.5-diff (`tumor_plus_differential_protection`) and V2.5-sigmoid (`tumor_plus_gap_sigmoid`).

The primary endpoint is GSE322563 validated-label AUC at the three Roth Fig. 5d target probes, evaluated under both HM450-intersect and native EPIC v2 ingest paths. **Because the validated set contains only three Roth Fig. 5d targets, AUC is interpreted as a rank-lift summary rather than an inferential discovery-performance estimate.** V2.5 axes place all three positives in the top percentile; a 10^6-draw random-triple null resolves the rank lift beyond the earlier 10^4 floor, while the paired V2.5-vs-Δβ comparison remains descriptive at `n_pos = 3`. The strongest empirical result is on GSE69914 tissue: a probe-level limma-style moderated-t DMR baseline is competitive on matched cell lines but falls to AUC 0.573 on tissue, while V2.5-sigmoid reaches AUC 0.862 on the same candidate mapping. A frozen whole-genome panel shows V2.5-sigmoid matches V2.5-diff's cell-line AUC within 0.002, avoids V2.5-diff's whole-genome low-`n` top-K tied bands (`tie_band@100 = 1` vs 421–1,493), and improves tissue AUC by +0.05 to +0.08.

The recommended probabilistic prioritization axis (hypothesis generation; not a discovery claim absent prospective wet-lab validation) is V2.5-sigmoid; V2.5-diff is retained for backward-compatible AUC parity,

V1 remains the package default for deterministic top-K behavior, and `tumor_only` remains the default `probabilistic_mode` enum value for compatibility. **V2.5-sigmoid is selected by post-repair sensitivity and whole-genome stress testing; its prospective ranking utility remains to be validated on newly generated labels or wet-lab follow-up.** This is a scoring-method and benchmarking contribution on public methylation data, not prospective editing validation. Whole-genome catalogs are SHA256-frozen and all benchmark artifacts are committed under `examples/`.

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1 · Background

1.1 ThermoCas9 is methylation-sensitive at the PAM

Roth et al. (2026) characterized *Geobacillus thermodenitrificans* T12 Cas9 (ThermoCas9) biochemically, structurally, and in human cell-line editing assays. The central result: cleavage efficiency is governed by the methylation state of the fifth position of the PAM (`5'-NNNN**C**GA-3'` or `5'-NNNN**C**CA-3'`), not by methylation within the protospacer. Methylated 5-methylcytosine (5mC) at the PAM abolishes binding; unmethylated PAMs are cleaved as usual. Cryo-EM structures at 2.2–2.8 Å resolved the protein contacts that discriminate methylated from unmethylated cytosine.

The practical consequence is that ThermoCas9 gives a natural mechanism to selectively edit in cells where a target locus is hypomethylated while sparing cells where the same locus is methylated. Roth demonstrated this in MCF-7 vs. MCF-10A at three breast-cancer loci: *EGFLAM* (control locus included in the validated target set), *ESR1*, and *GATA3*. The underlying methylation of the PAM cytosine at those loci was profiled on the Illumina Infinium MethylationEPIC v2 array and reported in Supplementary Figure 5d of the main paper alongside the editing results.

1.2 Target-site selection is a ranking problem

Given a reference genome, the ThermoCas9-compatible PAM sites on any chromosome number in the millions. To pick candidates for wet-lab follow-up, one wants a ranked list where the top scorers are *biologically coherent* — meaning the candidate's PAM cytosine is unmethylated in the disease cohort and methylated in the matched normal — *and* where the confidence of the underlying methylation measurement is part of the score. This is a classical ranking-with-covariates problem, and the natural scoring target is some decomposition of a probability of "therapeutic selectivity."

This paper describes the specific decomposition we ended up using after two earlier attempts failed on empirical cohorts.

1.3 Notation

For one candidate site on one cohort we carry six per-probe β summary statistics from the cohort's methylation array (`β_tumor_mean`, `β_tumor_q25`, `β_tumor_q75`, `β_normal_mean`, `β_normal_q25`, `β_normal_q75`), per-side sample counts (`n_tumor`, `n_normal`), and an `EvidenceClass` capturing the distance between the candidate's PAM cytosine and the nearest assayed CpG probe (`EXACT`/`PROXIMAL_CLOSE`/`PROXIMAL`/`REGIONAL`/`UNOBSERVED`) — nine inputs in total per candidate × cohort pair.

---

2 · The V2 misspecification

2.1 First-pass composite

The first-pass composite followed a standard decomposition:

```
p_therapeutic_selectivity = p_targ × p_prot × p_trust
```

- `p_targ = P(β_tumor < 0.30)` — probability that the tumor's PAM cytosine is unmethylated enough for ThermoCas9 to cleave. Estimated via a method-of-moments Beta fit to `(mean, IQR/1.349)` with the regularized incomplete beta computed in pure stdlib via Lentz's continued fraction; falls back to a piecewise-linear CDF when the Beta moments are ill-defined.
- `p_prot = P(β_normal > 0.50)` — probability that the normal's PAM cytosine is methylated enough to protect the normal cell.
- `p_trust` — confidence factor that scales with `EvidenceClass` (0.95 at EXACT, 0.15 at REGIONAL, 0.0 at UNOBSERVED) and saturates linearly in `min(n_tumor, n_normal)` up to `ramp_n = 30`.

2.2 Empirical failure of `p_prot`

We ran a factor ablation on a structural surrogate for the Roth MCF-7 vs. MCF-10A comparison: GEO series GSE77348, an independent 2016 breast cell-line methylation series on HM450 (not the Roth samples). With 1,687 gene-region-filtered "positives" at the Roth genes, the measured AUCs were:

axis	AUC (loose)	AUC (tight)
`v2_tumor_only` (p_targ × p_trust)	0.733	0.770
`p_targ` alone	0.683	0.717
V1 `final_score`	0.657	0.628
`naive_selectivity` (β_n − β_t)	0.608	0.571
`p_targ × p_prot`	0.503	0.488
`p_prot` alone	**0.384**	**0.343**

`p_prot` is anti-predictive. Multiplying it against `p_targ` destroys the signal. The cause is the semantics: `P(β_normal > 0.5)` assumes that the normal comparator *methylates the target*. That assumption holds for Roth's specific MCF-10A cell line at the three *ESR1* / *GATA3* / *EGFLAM* probes they chose to validate (genes that are silenced in non-tumorigenic mammary epithelium) but does not generalize to bulk adjacent-normal cohorts or even to MCF-10A at gene-body probes.

2.3 V2.4 — keep the composite, drop `p_prot` by default

The intermediate fix kept the composite but allowed users to opt in or out of the anti-predictive factor. Default changed to `tumor_only` = `p_targ × p_trust`. This improved global AUC but the top-100 collapsed onto "always-unmethylated" loci (candidates where both tumor *and* normal are low-methylated) — good AUC, unusable for target discovery. The deeper fix was needed.

---

3 · V2.5 · two gap-factor instances (V2.5-diff and V2.5-sigmoid)

Terminology note. "V2.5" names the *generation* of the compositional skeleton `p_targ × (gap factor) × p_trust`. Two gap-factor instances ship as selectable `probabilistic_mode` enum values in this tag:

- **V2.5-diff** = `tumor_plus_differential_protection`, the original V2.5 composite with `p_diff` as the gap factor. This section defines it.
- **V2.5-sigmoid** = `tumor_plus_gap_sigmoid`, a fixed-bandwidth sigmoid replacement of `p_diff`. Introduced and benchmarked in §5.2.1 + §5.2.2; the recommended probabilistic prioritization axis per the §5.2.2 whole-genome panel.

Where the prose says "shipped V2.5" without qualifier it historically means V2.5-diff (the axis under test in §5.1–§5.2 and the axis the scored JSONLs in `data/derived/scored_*.jsonl` were computed with); we keep the V2.5 / V2.5-diff / V2.5-sigmoid disambiguation explicit when a specific variant matters.

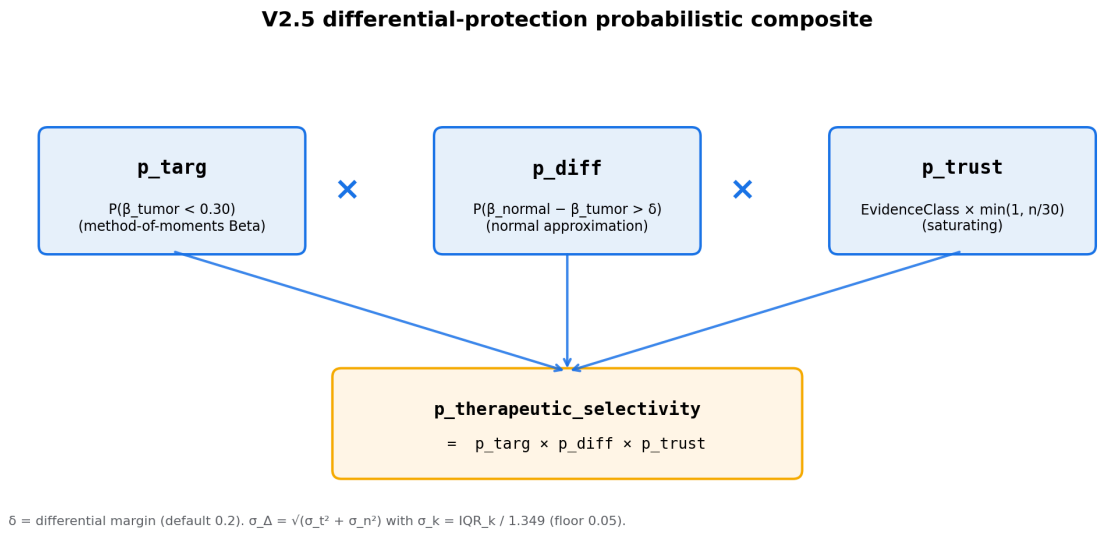

Figure 1. The V2.5-diff composite. Three factors multiply: `p_targ` (tumor unmethylated at the PAM cytosine), `p_diff` (differential methylation gap exceeds δ under a normal approximation on per-probe β summaries), and `p_trust` (evidence-class confidence, saturating in min sample count). The result is the *probability-scale selectivity score* (the stored scalar that downstream ranking consumes; stored in the `p_therapeutic_selectivity` field on every `ProbabilisticScore` record). The deprecated V2 `p_prot = P(β_normal > 0.5)` is replaced by `p_diff` to remove its static-threshold assumption about the normal side. V2.5-sigmoid swaps `p_diff` for a fixed-bandwidth sigmoid of the same (β_n − β_t − δ) shift, motivated by the §3.5 binding-rate finding and benchmarked in §5.2.1 / §5.2.2.

3.1 Formulation

Replace the threshold-based `p_prot` with a differential factor:

```
p_diff(δ) = P(β_normal − β_tumor > δ)
```

estimated under an independent-normal approximation:

```
σ_k   ≈ IQR_k / 1.349                (k ∈ {tumor, normal}; floor at 0.05)
σ_Δ²  = max(σ_t, floor)² + max(σ_n, floor)²
z     = (δ − (μ_n − μ_t)) / σ_Δ
p_diff = 1 − Φ(z) = 0.5 (1 − erf(z / √2))
```

with Φ the standard normal CDF. The composite becomes:

```
p_therapeutic_selectivity = p_targ × p_diff × p_trust
```

This makes no absolute-threshold claim about the normal side. It asks only what fraction of normal-vs-tumor β pairs sampled from the per-side empirical distributions would exceed δ, under independent normal approximations to those per-side distributions. (This is a *population-overlap* statistic, not a confidence interval on a cohort-level mean contrast — see §3.3.)

Calibration scope. `p_therapeutic_selectivity` is a *bounded probability-scale ranking score in [0, 1]*, not a calibrated probability of editing success at the candidate site. The composite is not validated against any prospective editing readout (no expected calibration error is reported, no reliability diagram is fit, no Platt/isotonic post-calibration is applied). What the bounded scale buys is multiplicative composability with downstream probabilistic inputs (target-mutation models, gRNA off-target probabilities, delivery-efficiency priors) — not a frequentist guarantee that "0.9" means "90% of 0.9-scored sites edit successfully." Read the score as a probability-scale ranking axis throughout.

3.2 Choice of δ

An offline δ sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 surrogate selected δ = 0.2 by joint optimization of AUC and `P@100` against the *pre-repair* (gene-universe) positives that were in use at the time of selection. Under that pre-repair tuning objective, larger δ values monotonically reduced both metrics — the factor was too strict to admit candidates with moderate but real differentials.

The post-repair sensitivity story is more nuanced and is reported in §5.3.2: against the Roth-validated label set, δ = 0.3 (and on some cohort × tier rows, δ = 0.4) marginally outperforms δ = 0.2 on the matched cell-line cohorts; tissue (GSE69914) prefers smaller δ. We do not retune δ post hoc against the post-repair labels, because doing so would invalidate the dev-cohort separation discipline of §4.0. δ = 0.2 is therefore retained as the shipped default; readers wanting the full sensitivity surface should consult §5.3.2.

The default is exposed as `differential_delta` in the cohort configuration. The δ used is recorded on every emitted `ProbabilisticScore` record so the margin in effect is recoverable from a single output row.

3.3 What `p_diff` is a probability over

`p_diff(δ) = P(β_normal − β_tumor > δ)` is a probability under the *biological β-heterogeneity* implied by per-cohort, per-probe quartiles — not a posterior probability of editing efficacy and not a confidence interval on a cohort-level mean contrast. The distribution being integrated is the one that would generate single-sample β values from a tumor-side population with quartiles (`β_tumor_q25`, `β_tumor_q75`) and a normal-side population with quartiles (`β_normal_q25`, `β_normal_q75`), under independent normal approximations on each side. Two consequences follow:

1. **The variance does not shrink with sample size.** Because the per-side σ is the IQR-derived population dispersion, not the standard error of a mean, `p_diff` does not become more confident as `n_tumor`/`n_normal` grow. The growing-`n` story for the composite is carried by `p_trust` (saturating at `n ≳ 30`), not by `p_diff` tightening.
2. **`p_diff` is therefore a population-overlap statistic, not a significance test.** It answers "what fraction of tumor-vs-normal β-pairs sampled from the cohort would exceed δ?" An alternative formulation that scales σ with `n` (e.g. SE on the mean difference, or a hierarchical model on per-side variance) would also be defensible and would give a different — and at low `n`, a much wider — distribution. We use the population-overlap form because (a) the comparison the user cares about for site selection is biological — "is this site differentially methylated *between cells*?" — not "does this cohort have enough power to call this site significant?"; and (b) the small-`n` power question is already absorbed by `p_trust`. Re-deriving the composite under an SE-on-mean variant is a documented future axis (§6.3).

This is a deliberate modeling choice and is the single most common point of confusion when reading `p_diff` cold.

3.4 Why a normal approximation instead of a difference-of-Betas

A proper Bayesian treatment would model β-values as Beta distributions and compute the exact difference distribution numerically. For the IQR ranges seen on HM450 / EPIC v2 arrays (~0.05–0.20), the normal

approximation is close to the correct answer and has the advantage of being a two-line stdlib closed form. A σ floor of 0.05 prevents σ_Δ from collapsing to zero at boundary β-values (very common for CpG islands methylated at 0 or 1). These tradeoffs are documented in the code; a downstream user who wants the exact difference-of-Betas path can swap it in.

The normal approximation is operationally adequate for the *ranking* use case but is not a calibrated probability of editing success. Two situations where the approximation is poor on its own terms but does not corrupt the ranking:

- **Boundary-β pile-ups.** β-distributions at CpG islands often pile up near 0 or 1; a single-mode normal centered at the per-side mean overstates symmetry and understates skew. The σ floor blunts the worst case (σ_Δ ⟶ 0) but does not restore the correct shape.
- **Bimodality across samples.** When tumor or normal samples carry two β sub-populations (e.g. methylation-class heterogeneity in primary tissue), the IQR-derived σ averages the two modes; the per-side normal then sits between the modes rather than on either. `p_diff` in this regime under-states the true population overlap.

Neither failure mode is silent: the §5.5 annotation pipeline flags candidates whose per-side β quartiles indicate strong bimodality. A future axis based on the exact difference-of-Betas or a two-component mixture would address both.

3.5 The low-n tied-band prediction

With the new factor isolated, we can predict before running any new cohort. `p_trust` is `EvidenceClass.base × min(1, min(n_t, n_n) / 30)`: piecewise-linear in min sample count up to `ramp_n = 30`, then a constant ceiling per evidence class. It is **never** continuous-valued in the real-line sense — at n ≥ 30 it saturates to a discrete-by- EvidenceClass value (0.95 at EXACT, 0.75 at PROXIMAL_CLOSE, etc.). What changes between the low-n and high-n regimes is which factor *dominates* the composite ordering:

- **Low n (n ≪ 30).** `p_trust` scales every same-evidence-class candidate by the same uniform fraction `(n / 30)`. Per-side IQRs are wide (or zero, hitting the `σ_floor`), pushing both `p_targ` and `p_diff` toward saturation (≈1) for any candidate with clean tumor hypomethylation and a large normal/tumor gap. The composite is effectively `1 × 1 × p_trust` for that whole subset, producing a large tied band at the top of the score distribution.
- **High n (n ≥ 30).** `p_trust` is constant per EvidenceClass. Per-side IQRs are realistic (HM450 ranges of 0.05–0.20). The composite ordering is now driven by the continuous variation in `p_targ × p_diff`, with `p_trust` only multiplying by a per- evidence-class constant. The previously-tied subset spreads out by its underlying `p_targ × p_diff` values.

The falsifiable prediction: the K = 100 tied band on V2.5-diff should be large at n=2/3 per side and shrink to near-1 at n ≳ 30 per side. §5.1 and §5.3 test this directly across cohorts ranging from n=2/2 to n=305/50.

The K = 100 large-band regime is what motivates the "top tied candidate class" framing in §6.1: on n=2/2 cohorts the visible top-20 is a 20-record window inside a 190- to 421-record tied band, not a stable rank ordering. Per-positive ranks (§5.1) and AUC (§5.1, §5.3) are the stable claims at low `n`.

Quartile convention on n=2 records. Per-side β quartiles are computed with Python's `statistics.quantiles(..., method="inclusive")` (R-7 quantile definition) and hard-clamped to the observed sample range at every probe, so q25 / q75 / mean are always within [min(β_samples), max(β_samples)]. On n = 2 samples per side this is deterministic (IQR = 0.5 × |β_max − β_min|) but weak as a population-dispersion estimate: the "IQR" it produces is a function of two observations, not a quantile of

an underlying distribution. Because the `σ_floor = 0.05` kicks in whenever IQR/1.349 < 0.05 (i.e. whenever the two observations differ by less than ~0.135), σ_floor dominates σ_Δ on most n = 2 records.

Quantified σ_floor binding rate per cohort (reproducible via `uv run scripts/sigma_floor_binding_rate.py`; denominator is records with both β means present):

cohort	σ_floor binds on either side	σ_floor binds on both sides
GSE322563 HM450 (n = 2/2)	**100.0%**	**99.9%**
GSE322563 native (n = 2/2)	**100.0%**	**99.9%**
GSE77348 (n = 3/3)	**100.0%**	**99.5%**
GSE69914 (n = 305/50 tissue)	65.1%	43.2%

On matched cell-line cohorts the σ_floor is the binding constraint on essentially every observed record (both sides). On tissue the true IQRs are wide enough that the σ_floor binds on roughly two thirds of records either-side, and on both sides for only ~43%. This is the mechanistic origin of both the low-`n` tied band (§5.3.1) and the §5.3.2 finding that tissue AUC prefers a larger σ_floor: on cell-line cohorts, σ_floor is determining σ_Δ almost everywhere, so changing its value re-shapes the whole composite.

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4 · Benchmark methodology

4.0 Analysis plan: primary endpoint, sensitivity, boundary

The design separates tuned components from evaluation. It is not a third-party preregistration, but the ordering is recoverable from git: `differential_delta = 0.2` was fixed at commit `aece3de` on the GSE77348 surrogate before GSE322563 was ingested.

Analysis role	Cohort / labels	Interpretation
Tuned component	GSE77348, pre-repair gene-universe labels	δ-development cohort; later AUC is tuned-on supporting evidence.
Primary endpoint	GSE322563 validated Roth probes (`n_pos = 3`)	Independent paper-comparable rank-lift test; read with per-positive ranks.
Sensitivity	narrow ±50 bp / wide ±500 bp labels, native EPIC v2 ingest	Stability checks without retuning.
Tissue behavior	GSE69914, n = 305 / 50	Tests high-`n` tissue behavior and tied-band dissolution, not Roth biology.
Boundary	GSE68379 × GSE69914 normal	Label-transportability stress test; never pooled with discovery results.

4.1 Cohorts

Four cohorts, all public HM450 or EPIC v2 methylation arrays:

cohort	regime	n tumor / normal	platform	role
GSE322563	Roth actual samples	2 / 2	EPIC v2	paper-comparable biology
GSE77348	MCF-7/ MCF-10A surrogate (DNMT3B paper)	3 / 3	HM450	δ development cohort (tuned-on supporting evidence)
GSE69914	primary breast tissue	305 / 50	HM450	high-n tissue validation
GSE68379	Sanger GDSC cell-line panel × GSE69914 healthy normal	52 / 50	HM450	orthogonal (see §5.4)

All cohort builders live in `scripts/build_*.py` and emit the same `LocalSummaryBackend` format (`tumor_summary.tsv`, `normal_summary.tsv`, `PROVENANCE.md`). GSE322563 uses the Roth EPIC v2 β matrix; its HM450- intersect path strips beadchip suffixes from EPIC v2 probe IDs and retains 80.7% of probes, while the native EPIC v2 path is reported as a sensitivity. Per-probe β values at the three Roth target positions match the published values to two decimals. GSE77348 is restricted to untreated replicates, GSE69914 excludes adjacent-normal and BRCA1-carrier arms, and GSE68379 is used only as a cross-series boundary case.

4.2 Positives (label repair)

Our first benchmark used "gene-universe" positives: HM450 probes in the neighborhood of Roth's validated gene symbols (~970 tight, 1,687 loose, on chr5/6/10). A diagnostic on GSE322563 revealed this label set is noisy: only 23% of Roth-gene probes have $\beta_n - \beta_t > 0.2$ on the actual Roth samples. Most are gene-universe members, not validated target loci.

We rebuilt the positives from Roth Supplementary Fig. 5d which names three exact MCF-7 / MCF-10A validated target coordinates in hg38. These were lifted to hg19 via Ensembl REST `/map/human/GRCh38/.../GRCh37`:

target	gene	hg38	hg19 (lifted)	β shift (hg19 vs hg38)
EGFLAM T11	EGFLAM	chr5:38,258,842	chr5:38,258,944	+102 bp
ESR1 T17	ESR1	chr6:151,690,043	chr6:152,011,178	+321,135 bp
GATA3 T18	GATA3	chr10:8,045,425	chr10:8,087,388	+41,963 bp

Three positives files result, at increasing granularity:

- `positives_roth_validated.txt` (n = 3): closest NNNNCGA candidate per Roth target.
- `positives_roth_narrow.txt` (n = 28): NNNNCGA candidates within ±50 bp of each Roth target.
- `positives_roth_wide.txt` (n = 142): NNNNCGA candidates within ±500 bp of each Roth target.

4.3 Score axes

Seven axes are benchmarked. For naming conventions see §7 Methods (canonical axis-naming table):

- **Δβ-only** (`naive_selectivity`) — `β_normal_mean − β_tumor_mean`. The literature-naive baseline: rank by the raw methylation gap with no uncertainty propagation or evidence weighting. Reported up front so the value of the additional factors is auditable against the simplest possible scoring axis.
- **Δβ_z** — `(β_n − β_t) / sqrt(σ_t² + σ_n²)` using the same IQR-derived σ as V2.5-diff's `p_diff`. An uncertainty-aware effect-size ranker built from the same per-side dispersion signal; confirms that V2.5-diff's advantage is not just an alias for σ-normalized Δβ.
- **V1** — `final_score = sequence × selectivity × confidence − heterogeneity_penalty − low_coverage_penalty`. Deterministic, continuous-valued, tie band always 1.
- **V2** (`tumor_only`) — `p_targ × p_trust`. Retained for audit; default in v0.4.0.
- **V2.5-diff** (`tumor_plus_differential_protection`) — `p_targ × p_diff × p_trust` with δ = 0.2 and σ_floor = 0.05. The original V2.5 composite; retained in this tag for backward compatibility and AUC parity on cell-line cohorts.
- **V2.5-sigmoid** (`tumor_plus_gap_sigmoid`) — `p_targ × sigmoid((β_n − β_t − δ) / σ_fixed) × p_trust` with δ = 0.2 and σ_fixed ≈ 0.0707. The **recommended probabilistic prioritization axis** on every non-boundary cohort shape tested (§5.2.2 whole-genome panel); ships as a first-class `probabilistic_mode` enum value in this tag.
- **limma-style moderated-t** — sample-level probe-level Smyth (2004) empirical-Bayes moderated `t`-statistic, candidate-mapped via the nearest probe assigned in `observation.probe_id`. Pure-Python reimplementation of the variance-prior math (not a byte-identical R `minfi` + `limma` Bioconductor workflow); a canonical R parity check is on the §6.3 follow-up list. Included as the methods-journal-standard DMR comparator; §5.1 + §5.2.2 report AUC + tie-band alongside the other axes.

limma-style moderated-t baseline. Standard DMR tools rank CpG probes or regions, not per-PAM candidates, so we compute a probe-level Smyth (2004) empirical-Bayes moderated `t` statistic from sample-level β matrices and map each candidate to its nearest probe via the same `EvidenceClass` assignment. This is a limma-style statistic implemented in pure Python, not a byte-identical `minfi` + `limma` Bioconductor workflow; an R parity check is future work. It is reported in §5.1 and §5.2.2 because it is the methods-reviewer baseline.

4.4 Platform harmonization and catalog scope

Two facts constrain interpretation of the reported numbers and belong in the methodology, not as a footnote:

- **Platform.** GSE322563 is EPIC v2; the other cohorts are HM450. §5.1 uses the HM450-intersect path for cross-cohort comparability and §5.2 reports a native EPIC v2 sensitivity. The validated-label V2.5-diff AUC changes by only 0.003 (0.990 ⟶ 0.986); V1 changes more, but the matched-cell-line ordering V2.5-diff > V1 and V2.5-diff > Δβ-only holds on every tier × path row.
- **Catalog scope.** Three frozen probe-window catalogs are used across §5, each with its own SHA256-committed provenance file:
  - **chr5 / chr6 / chr10 HM450 (2.98M candidates)** — `catalog_hg19_chr5_6_10.jsonl`. The primary benchmark catalog used by §5.1 (primary endpoint) and §5.2 (label granularity, tie-band diagnostics, factor ablation, σ_floor and δ sweeps, Δβ-only baseline). Chosen because the three chromosomes carry the Roth Fig. 5d validated targets (EGFLAM on chr5, ESR1 on chr6, GATA3 on chr10) and give a realistic test bed.

- **Whole-genome HM450 (19.8M candidates, SHA256 `d20661c5…`)** — `catalog_hg19_wg.jsonl`. The genome-wide gating catalog used by §5.2.2 for GSE322563 HM450 / GSE77348 / GSE69914. Frozen provenance: `data/derived/catalog_hg19_wg.PROVENANCE.md`.
- **Whole-genome native EPIC v2 (35.4M candidates, SHA256 `39df8f0f…`)** — `catalog_hg19_wg_epic_v2.jsonl`. The GSE322563 native-path genome-wide catalog used by §5.2.2. Frozen provenance: `catalog_hg19_wg_epic_v2.PROVENANCE.md`.

Every AUC / `P@K` / tie_band number in §5 names its catalog explicitly. §5.1 and §5.2 are chr5/6/10-scope; §5.2.2 is whole-genome-scope; §5.3 (tissue behavior) carries both the chr5/6/10 and WG rows. Extending the chr5/6/10 primary endpoint to random-chromosome-matched controls is the one remaining untested denominator dimension (§6.3).

4.5 Tie-band handling

Earlier iterations of the benchmark sorted candidates by score alone, with input-order as the implicit tie-break. On cohorts with low n, the score distribution's top can sit inside a tied band of 190–11,000 records, making the reported `P@K` a function of serialization order rather than of the scorer.

The shipped benchmark contract:

1. Sort primary by score descending, secondary by `candidate_id` ascending (`evaluate_ranking`'s explicit `tie_break_policy = "candidate_id_asc"`).
2. Report `tie_band_size_at_k`: number of records tied at the K-th position's score. When > 1, top-K membership is partially determined by the secondary key.
3. Report `precision_at_k_min` and `precision_at_k_max` (plus recall counterparts) — adversarial tie-break bounds over the tied band. The worst-case pushes every tied-band positive out of top-K; the best-case pulls them in. When `tie_band_size_at_k == 1` both bounds collapse to the observed `precision_at_k`.

These three fields are on every emitted `BenchmarkResult` JSONL row.

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5 · Results

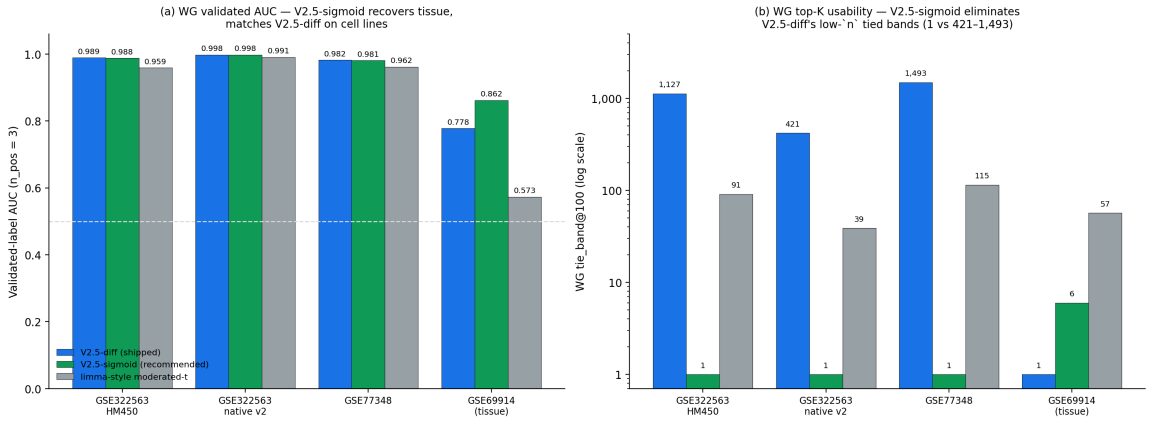

Figure 2 — Final-method summary. Whole-genome validated-label performance (n_pos = 3 Roth Fig. 5d target probes) for the recommended probabilistic prioritization axis (V2.5-sigmoid) against the V2.5-diff predecessor and the limma-style moderated-`t` DMR baseline, on the four primary cohorts. **(a)** WG AUC: V2.5-sigmoid matches V2.5-diff on the three matched cell-line cohorts within 0.001 (0.989 vs 0.988, 0.998 vs 0.998, 0.982 vs 0.981) and beats it on tissue (0.862 vs 0.778); the limma-style baseline is competitive on cell lines (0.959 / 0.991 / 0.962) but **collapses to near-random on tissue** (0.573). **(b)** WG `tie_band@100` on log

scale: V2.5-sigmoid eliminates the **421–1,493 record tied bands** that make V2.5-diff's whole-genome top-K unactionable on low-n cell-line cohorts, holding `tie_band@100 = 1` on every matched cell-line cohort and `= 6` on tissue. The limma-style baseline sits at intermediate tie-band values (39–115). The supplementary historical 4-axis × 3-tier sensitivity figure (`docs/figures/fig2_supp_historical_sensitivity.png`) is retained for auditability of the V2.5-diff-era sensitivity story. Numbers committed in `examples/genome_wide_panel.tsv` (V2.5-diff, V2.5-sigmoid) and `examples/limma_cross_cohort_panel.md` (limma-style) at this tag.

5.1 Primary endpoint — matched cell-line AUC at validated Roth probes

Primary endpoint (independent): AUC at `positives_roth_validated.txt` (n = 3 Roth Fig. 5d target probes, hg38 ⟶ hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples, EPIC v2). δ was not tuned on this cohort; the label set was not derived from it.

Supporting evidence (development cohort for δ): AUC on **GSE77348** (MCF-7 / MCF-10A on HM450, the DNMT3B-paper surrogate). Same matched cell-line biology, different platform and laboratory. Reported here because same-axis ordering on a second lab's MCF-7/MCF-10A is informative, but with the caveat that δ was selected on this cohort (§4.0) so its AUC is tuned-on, not independent.

axis	**GSE322563** (primary, n_pos=3, n_neg=2,979,994)	GSE77348 (δ-tuned, n_pos=3, n_neg=2,979,994)
Δβ-only (`β_n - β_t`)	0.974	0.972
Δβ_z (uncertainty-aware Δβ)	0.940	0.956
limma-style moderated-`t` (probe-level ⟶ candidate-mapped)	0.959	0.962
V1 `final_score`	0.821	0.968
V2 `tumor_only`	0.928	0.912
V2.5-diff (`tumor_plus_differential_protection`)	0.990	0.982
V2.5-sigmoid (recommended)	**0.989**	**0.982**

Reading the axis grid. Δβ-only (raw mean gap) is already a strong baseline — it lifts the validated positives to AUC 0.974 on the primary cohort. The uncertainty-aware Δβ_z baseline (`(β_n - β_t) / sqrt(σ_t² + σ_n²)`, using the *same* IQR-derived σ as V2.5′s `p_diff` denominator; reproducible via `uv run scripts/delta_beta_z_baseline.py`) lifts to AUC 0.940 — slightly *below* raw Δβ-only because the σ-based denominator penalizes positives with wide IQRs. V2.5-diff adds **+0.016** over Δβ-only and **+0.050** over Δβ_z on the primary endpoint; on GSE77348 the corresponding margins are +0.010 and +0.026. V2.5-sigmoid (the recommended axis — see §5.2.2) is AUC-equivalent to V2.5-diff on both cohorts within 0.001 (0.989 vs 0.990 on GSE322563, 0.982 vs 0.982 on GSE77348), consistent with the §5.2.1 ablation finding that `p_diff`'s per-site structure and the fixed-bandwidth sigmoid are interchangeable on cell-line cohorts where σ_floor binds ~99% of records; the reason V2.5-sigmoid is nonetheless recommended over V2.5-diff is top-K usability at WG scale on cell-line cohorts, not AUC on chr5/6/10 — see §5.2.2 for the `tie_band@100` numbers. These margins are small but consistent across the matched cell-line cohorts and label granularities tested (§5.2). The `p_targ` (tumor-side unmethylation) and `p_trust` (evidence / sample saturation) factors of both V2.5

variants are pulling real weight on top of the per-side dispersion signal — V2.5 is not just a re-skinned Δβ_z — but the honest framing is that the shipped baselines (especially raw Δβ-only) already do most of the rank-lift work on matched cell-line cohorts; V2.5′s additional contribution is the tie-band-honest top-K reporting (§4.5) and the probability-scale composition with future factors, not the AUC point alone.

On the independent primary endpoint (GSE322563 validated), V2.5-diff is the highest-AUC axis — +0.17 over V1 and +0.06 over the deprecated V2 `tumor_only` composite. On the development cohort (GSE77348 validated), V2.5-diff is similarly the highest-AUC axis (+0.01 over V1, +0.07 over `tumor_only`); this is consistent with the primary- endpoint result but is not an independent replication, since the same cohort selected δ = 0.2 pre-repair-labels (§4.0).

5.1.1 Per-positive ranks of the n = 3 validated targets

AUC summarizes the rank lift across the three validated positives, but with `n_pos = 3` it is highly sensitive to the rank of each one. The table below gives the per-positive rank, percentile rank, score, and per-side β values on each cohort × axis. (Full grid including GSE69914, all four axes including Δβ-only, is at `examples/validated_positive_ranks.md`; reproduce with `uv run scripts/validated_positive_ranks.py`.)

GSE322563 HM450 (primary, N=2,979,997 candidates):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5-diff rank (%ile)	Δβ-only rank (%ile)
ESR1	0.07	0.94	0.87	**844** (99.97%)	**2,575** (99.91%)	4,742 (99.84%)
EGFLAM	0.02	0.49	0.48	839,106 (71.84%)	64,433 (97.84%)	67,620 (97.73%)
GATA3	0.03	0.31	0.28	755,885 (74.64%)	24,083 (99.19%)	159,932 (94.63%)

GSE77348 (development cohort for δ, N=2,979,997):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5-diff rank (%ile)	Δβ-only rank (%ile)
ESR1	0.03	0.87	0.84	**1,618** (99.95%)	**3,965** (99.87%)	8,733 (99.71%)
EGFLAM	0.005	0.34	0.33	257,041 (91.37%)	124,925 (95.81%)	178,663 (94.01%)
GATA3	0.015	0.59	0.58	31,061 (98.96%)	**28,325** (99.05%)	61,254 (97.94%)

Reading the table. *ESR1* drives the AUC story: it is in the top ~0.05% of the score distribution under V2.5-diff on both matched cell-line cohorts, with V1 also placing it well. *EGFLAM* and *GATA3* sit in the top 1–3% under V2.5-diff — meaningful rank lift over V1, but **outside the top 100 on every axis tested**. None of the three validated positives appears in any cohort's top-20 (§5.5). This is the operational meaning of "rank lift on n=3": V2.5-diff reliably places known positives in the top percentile of millions of candidates, but on n=2/2 cohorts the top tied band at K=100 is large enough that the validated positives sit *behind* a 190- to 421-record block of equally-scored candidates.

The ESR1-dominance also implies AUC is fragile: if ESR1 alone moved out of the top 10,000 on GSE322563, the AUC would drop from 0.990 to roughly the average of the other two positives' percentiles ≈ 0.985 (still high), but the AUC *gap* between V2.5-diff and Δβ-only would also compress.

5.1.2 Descriptive AUC uncertainty

At `n_pos = 3`, inferential AUC intervals are coarse and dominated by the rank of each positive. We therefore report a 1,000,000-draw random-triple null: draw three random candidates without replacement, compute AUC under the same mid-rank tie handling, and estimate `Pr(AUC_random_triple ≥ AUC_observed)`.

cohort	axis	observed AUC	random-triple null 2.5–97.5%	one-sided p (Pr ≥ obs)
GSE322563 HM450	V1 final_score	0.821	[0.166, 0.817]	2.2×10^{-2}
GSE322563 HM450	Δβ-only	0.974	[0.181, 0.823]	8.6×10^{-5}
GSE322563 HM450	**V2.5 diff**	**0.990**	[0.338, 0.802]	$\mathbf{4 \times 10^{-6}}$
GSE322563 native v2	V1 final_score	0.933	[0.206, 0.823]	1.3×10^{-3}
GSE322563 native v2	Δβ-only	0.961	[0.177, 0.823]	3.1×10^{-4}
GSE322563 native v2	**V2.5 diff**	**0.986**	[0.313, 0.817]	$\mathbf{1.5 \times 10^{-5}}$
GSE77348	V1 final_score	0.968	[0.261, 0.823]	1.6×10^{-4}
GSE77348	Δβ-only	0.972	[0.177, 0.823]	1.1×10^{-4}
GSE77348	**V2.5 diff**	**0.982**	[0.176, 0.823]	$\mathbf{2.8 \times 10^{-5}}$

V2.5-diff is the strictest listed axis on every row and sits below the old 10^4-draw resolution floor, but this is an against-random check, not a superiority test between axes. The direct paired comparison with Δβ-only is descriptive (§5.1.3). Full output: `examples/auc_uncertainty_1e6.md`; the negative bootstrap is omitted from the body because holding the positives fixed makes the spread mechanically collapse to ±0.001.

5.1.3 Paired per-positive rank lift: V2.5-diff vs Δβ-only

The paired comparison with Δβ-only on the same three positives (`uv run scripts/paired_rank_lift_v25_vs_dbeta.py`):

cohort	*ESR1* Δβ-only rank / V2.5-diff rank	*EGFLAM* Δβ / V2.5-diff	*GATA3* Δβ / V2.5-diff	positives where V2.5-diff > Δβ-only
GSE322563 HM450	4,742 / 2,575	67,620 / 64,433	159,932 / 24,083	**3 / 3**
GSE322563 native v2	9,085 / 5,746	185,641 / 158,502	422,569 / 50,096	**3 / 3**
GSE77348	8,733 / 3,965	178,663 / 124,925	61,254 / 28,325	**3 / 3**

V2.5-diff ranks all three positives above Δβ-only on every matched cell-line row. The largest lift concentrates at *GATA3*, the positive with the smallest raw β gap, consistent with the composite helping most where naive Δβ is weakest. The sign-flip paired null has only 2^3 patterns and floors at one-sided $p = 0.125$ when all three signs agree, so this is descriptive 3-for-3 directional evidence, not an inferential superiority claim.

5.2 Sensitivity analyses: label granularity and P@K intervals

Label granularity (AUC). Stability of the primary-endpoint ordering under weaker label definitions (narrow ±50 bp, wide ±500 bp):

cohort	label set	n_pos	n_neg	Δβ-only	V1 tumor_only	V2.5-diff	V2.5-diff – Δβ-only
GSE322563	validated	3	2,979,994	0.974	0.821	0.928	**0.990**	+0.016
GSE322563	narrow	28	2,979,969	0.912	0.884	0.886	**0.942**	+0.030
GSE322563	wide	142	2,979,855	0.844	0.768	0.871	**0.910**	+0.066
GSE77348	validated	3	2,979,994	0.972	0.968	0.912	**0.982**	+0.010
GSE77348	narrow	28	2,979,969	0.964	0.969	0.911	**0.983**	+0.019
GSE77348	wide	142	2,979,855	0.910	0.931	0.887	**0.949**	+0.039

V2.5-diff remains the highest-AUC axis on both cohorts at every label granularity. No retuning of `δ` between rows.

P@K intervals. On `n = 2 / 2` and `n = 3 / 3` cohorts the score distribution at K = 100 sits inside a tied band (§3.5); P@100 is reported as an adversarial interval `[min, max]` per §4.5. On GSE322563 narrow labels:

axis	P@100 observed	P@100 min	P@100 max	tie_band@100
V2.5 differential	0.000	0.000	0.020	190
tumor_only	0.000	0.000	0.020	10,005
V1 final_score	0.020	0.020	0.020	1

V2.5′s interval is [0.000, 0.020] — under the deterministic `candidate_id` ascending tie-break, zero narrow-positives land in top-100; under the best-case adversarial tie-break inside the 190-tied band, 2 of 28 could. V1′s interval collapses to the observed value because V1′s score is continuous and `tie_band = 1`. `tumor_only`'s tied band (10,005) spans two orders of magnitude more of the score distribution than V2.5′s; top-K on `tumor_only` is not a meaningful quantity on this cohort.

Native EPIC v2 vs HM450-intersect on GSE322563. §4.4 describes the harmonization shortcut: GSE322563 EPIC v2 probe IDs are stripped of beadchip-design suffixes and intersected with the HM450 universe (80.7% retention). To verify the shortcut is not distorting the primary endpoint, we ran the full pipeline a second time against a native EPIC v2 catalog (5.22M candidates vs the HM450 catalog's 2.98M):

label set	n_pos	n_neg (HM450 / native)	axis	HM450-intersect AUC	native EPIC v2 AUC	Δ
validated	3	2,979,994 / 5,224,079	Δβ-only	0.974	0.961	−0.013
validated	3	2,979,994 / 5,224,079	V1	0.821	0.933	+0.112
validated	3	2,979,994 / 5,224,079	tumor_only	0.928	0.936	+0.008
validated	3	2,979,994 / 5,224,079	**V2.5 differential**	**0.990**	**0.986**	**−0.003**
narrow	28	2,979,969 / 5,224,054	Δβ-only	0.912	0.956	+0.044
narrow	28	2,979,969 / 5,224,054	V1	0.884	0.938	+0.054
narrow	28	2,979,969 / 5,224,054	tumor_only	0.886	0.933	+0.046
narrow	28	2,979,969 / 5,224,054	V2.5 differential	0.942	0.983	+0.040
wide	142	2,979,855 / 5,223,940	Δβ-only	0.844	0.865	+0.021
wide	142	2,979,855 / 5,223,940	V1	0.768	0.855	+0.087
wide	142	2,979,855 / 5,223,940	tumor_only	0.871	0.916	+0.044
wide	142	2,979,855 / 5,223,940	V2.5 differential	0.910	0.945	+0.035

The V2.5-diff primary-endpoint AUC moves by 0.003 (well within the tied-band noise floor on this n=2/2 cohort). V1 gains the most under native EPIC v2 — its continuous score ranks the validated targets higher relative to easy-negative candidates added by the larger catalog. The relative axis ordering (V2.5-diff > V1)

is preserved across both ingest paths. The HM450-intersect shortcut is therefore not materially distorting the headline V2.5-diff claim. Tied bands grow modestly with the larger catalog (V2.5-diff: 190 ⟶ 421; tumor_only: 10,005 ⟶ 14,914; V1 stays at 1).

5.2.1 Factor ablation: does `p_diff` do the work, or does the floor?

The §3.5 binding-rate diagnostic shows that `σ_floor = 0.05` is the binding constraint on σ_Δ for essentially every observed record on matched cell-line cohorts (99.5–99.9% both-sides). That raises an obvious reviewer question: if σ_floor is determining σ_Δ almost everywhere on cell lines, is `p_diff`'s per-site σ-adaptive structure actually adding anything, or is the composite effectively `p_targ × sigmoid((Δβ − δ) / σ_fixed) × p_trust` with a constant bandwidth?

We replace `p_diff` with two simpler gap factors, keeping `p_targ`, `p_trust`, and δ = 0.2 unchanged (reproducible via `uv run scripts/factor_ablation_vs_sigmoid.py`):

- **sigmoid**: `sigmoid((Δβ − δ) / σ_fixed)` with σ_fixed = √2 × σ_floor ≈ 0.0707 — the σ_Δ V2.5 sees when the floor binds on both sides (the modal case, §3.5).
- **hard threshold**: `1[Δβ > δ]` — tests whether smoothness matters at all.

AUC at the n = 3 Roth-validated positives:

cohort	V2.5-diff (full `p_diff`)	sigmoid ablation	hard-threshold ablation
GSE322563 HM450	0.990	0.989	0.990
GSE322563 native v2	0.986	0.986	0.988
GSE77348	0.982	0.982	0.986
GSE69914 (tissue)	**0.773**	**0.864**	0.497

Two findings follow. On matched cell lines, `p_diff`'s per-site structure is not doing the AUC work: all three gap factors land within 0.001 AUC because σ_floor binds on essentially every record (§3.5). On tissue, the fixed-bandwidth sigmoid beats V2.5-diff by +0.09 AUC while the hard threshold collapses to chance. Smoothness matters on tissue; per-site σ adaptation does not help this label set.

Bandwidth robustness of the tissue finding (§5.2.1 single-point ablation used σ_fixed = √2 · σ_floor ≈ 0.0707; reproducible via `uv run scripts/sigmoid_bandwidth_sweep.py`):

cohort	σ_fixed=0.05	σ_fixed≈0.0707	σ_fixed=0.10	σ_fixed=0.15	shipped V2.5
GSE322563 HM450	0.990	0.989	0.989	0.977	0.990
GSE322563 native v2	0.986	0.986	0.985	0.976	0.986
GSE77348	0.982	0.982	0.979	0.965	0.982
GSE69914 (tissue)	**0.862**	**0.864**	**0.830**	**0.821**	**0.773**

Every tested σ_fixed beats V2.5-diff on tissue (+0.05 to +0.09 AUC); on matched cell lines the sigmoid family stays within 0.02 AUC except at the widest bandwidth. The tissue result is therefore not a one-point artifact and motivates V2.5-sigmoid as the recommended gap factor.

5.2.2 Genome-wide panel: shipped V2.5 vs `gap_sigmoid` across all four cohorts

The §5.2.1 ablation and bandwidth sweep are on the chr5/6/10 catalog (N = 2.98M candidates on HM450, 5.22M on native EPIC v2). A reviewer-salient concern is whether the findings survive when the negative universe expands to whole-genome. To resolve this we built two frozen whole-genome probe-window catalogs and ran the same gap-factor ablation on GSE322563 (both HM450 and native EPIC v2 paths), GSE77348, and GSE69914 against them:

- **HM450 WG catalog** — `catalog_hg19_wg.jsonl`, 19,787,820 candidates, SHA256 `d20661c5…` (provenance at `data/derived/catalog_hg19_wg.PROVENANCE.md`).
- **Native EPIC v2 WG catalog** — `catalog_hg19_wg_epic_v2.jsonl`, 35,380,431 candidates, SHA256 `39df8f0f…` (provenance at `data/derived/catalog_hg19_wg_epic_v2.PROVENANCE.md`). Built from a whole-genome re-run of `scripts/build_epic_v2_probes.py` (929,801 GPL33022 probes, hg38 → hg19).

Reproducible via `uv run scripts/genome_wide_tissue_gating.py` per cohort + `uv run scripts/genome_wide_panel.py` to aggregate. The committed panel is at `examples/genome_wide_panel.md`.

Cross-cohort AUC table at validated labels (n_pos = 3):

cohort	N_wg	V2.5 WG	gap_sigmoid σ=0.05 WG	gap_sigmoid σ=0.0707 WG	gap_sigmoid σ=0.10 WG
GSE322563 HM450	19.8M	0.989	0.989	0.988	0.988
GSE322563 native v2	35.4M	0.998	0.998	0.998	0.998
GSE77348	19.8M	0.982	0.982	0.981	0.979
GSE69914 (tissue)	19.8M	**0.778**	**0.861**	**0.862**	0.825

Cross-cohort `tie_band@100` table at validated labels (WG only):

cohort	V2.5 WG	gap_sigmoid σ=0.05	gap_sigmoid σ=0.0707	gap_sigmoid σ=0.10
GSE322563 HM450	**1,127**	1	1	1
GSE322563 native v2	**421**	1	1	1
GSE77348	**1,493**	1	1	1
GSE69914 (tissue)	1	4	6	6

The WG panel resolves three questions. First, matched-cell-line AUC is denominator-robust: all axes remain within 0.002 of the chr5/6/10 rows. Second, V2.5-diff's low-`n` top-K degeneracy appears only at scale:

`tie_band@100` expands to 421–1,493 records, while gap_sigmoid stays at 1 on all matched cell-line WG cohorts. Third, the tissue gap_sigmoid advantage survives the WG denominator (+0.05 to +0.08 AUC) with single-digit tie bands. P@100 remains zero for the three Roth positives at WG scale; the stable claim is rank lift, not top-100 recovery.

Does `gap_sigmoid` on tissue need δ = 0.1 (per §5.3.2′s `p_diff` finding) or δ = 0.2 (the shipped default)? §5.3.2 showed `p_diff` on tissue prefers δ = 0.1 over δ = 0.2 by +0.05 AUC. We re-ran the WG tissue gating at δ = 0.1 to check whether that property transfers to `gap_sigmoid`. Result (`examples/gse69914_wg_gating_delta_0_1.md`):

σ_fixed	gap_sigmoid AUC, δ = 0.2	gap_sigmoid AUC, δ = 0.1	Δ (δ=0.1 − δ=0.2)
0.05	0.861	0.865	+0.004
0.0707	0.862	0.831	**−0.031**
0.10	0.825	0.821	−0.004

The `p_diff` δ = 0.1 tissue advantage does not transfer cleanly to gap_sigmoid; at the default σ_fixed ≈ 0.0707, δ = 0.1 drops by 0.031 AUC. δ = 0.2 is therefore retained as the cross-mode default.

Together, these tables support the current recommendation: gap_sigmoid is uniformly equal-or-better than V2.5-diff across every tested non-boundary cohort × denominator combination, with AUC parity on matched cell lines, higher AUC on tissue, and far smaller WG top-K tied bands.

Cross-cohort comparison against the limma-style moderated-t DMR baseline (§4.3 exclusion-rationale paragraph; cross-cohort panel at `examples/limma_cross_cohort_panel.md`):

cohort	limma-style moderated-t AUC	V2.5-diff AUC	V2.5-sigmoid AUC	limma tie@100	V2.5-sigmoid tie@100
GSE322563 HM450	0.959	0.989	0.988	91	1
GSE322563 native v2	0.991	0.998	0.998	39	1
GSE77348	0.962	0.982	0.981	115	1
GSE69914 (tissue)	**0.573**	0.778	**0.862**	57	6

limma-style moderated-t is a credible matched-cell-line DMR baseline but collapses to near-random on tissue, where V2.5-sigmoid reaches 0.862. That contrast is the clearest evidence that the compositional wrapper (`p_targ × p_trust`) is doing work a pure probe-level differential statistic cannot. Reproducible artifacts: `examples/limma_cross_cohort_panel.md` and `examples/genome_wide_panel.md`.

5.3 Tissue-cohort behavior — GSE69914 (high-n, tissue biology)

GSE69914 (n = 305 / 50 primary breast vs. healthy donor tissue) tests whether the low-n tied-band behavior predicted in §3.5 dissolves at `n ≫ 30`:

label set	n_pos	n_neg	Δβ-only	Δβ_z	V1	tumor_only	V2.5	tie_band@100
validated	3	2,979,994	0.591	0.504	0.660	**0.803**	0.773	**2**
narrow	28	2,979,969	0.477	—	0.539	**0.843**	0.711	2
wide	142	2,979,855	0.435	—	0.435	**0.874**	0.726	2

On tissue, both Δβ-based baselines collapse to near-random or below (Δβ-only 0.435–0.591; Δβ_z 0.504 at validated). V1 tracks Δβ-only closely (0.435–0.660). The V2.5-generation tissue story is therefore a recovery *over both Δβ baselines*: V2.5-diff reaches +0.18 over Δβ-only at validated and +0.27 over Δβ_z, not a modest margin as on cell lines. `tumor_only` has a higher raw AUC (0.803–0.874) but its K = 100 tied band makes its top-K unusable. **V2.5-sigmoid** — which also ships in this tag as `tumor_plus_gap_sigmoid` and is the recommended probabilistic axis (§6.1) — **improves on V2.5-diff's tissue AUC by a further +0.05 to +0.09 across the bandwidth family tested in §5.2.1 (0.821–0.864 vs V2.5-diff's 0.773)**, at both the chr5/6/10 and whole-genome probe-window denominators (§5.2.2). The table below reports V2.5-diff's cross-tier tissue AUC; V2.5-sigmoid is the cross-mode improvement that the §5.2 follow-ups surface directly.

Two findings:

1. **The tied-band prediction holds.** V2.5′s tied band at K = 100 shrinks from 190 at `n = 2 / 2` ⟶ 299 at `n = 3 / 3` ⟶ 2 at `n = 305 / 50`, with no hyperparameter changes between cohorts. At `n ≥ 30`, `p_trust` saturates at the EvidenceClass ceiling (0.95 for EXACT) — a per-class *constant*, not a continuous function of n. The tied band shrinks not because `p_trust` becomes continuous-valued but because `p_targ × p_diff` are no longer pushed to 1 by wide-IQR saturation and the σ floor; with realistic-width IQRs (0.05–0.20) the composite's ordering is driven by continuous variation in those two factors while `p_trust` just multiplies by the class constant.
2. **The mode ordering flips on tissue biology.** `tumor_only` wins raw AUC (+0.03 to +0.15 over V2.5-diff / V2.5-sigmoid), but `tumor_only`'s tied band is 6,540 at K = 100 — its top-K is not usable. Both V2.5 variants have interpretable top-K on tissue (V2.5-diff `tie_band@100 = 2` chr5/6/10 / = 1 WG; V2.5-sigmoid `tie_band@100 ≤ 6` across the tested bandwidth family), and V2.5-sigmoid additionally wins on tissue AUC by +0.05 to +0.08 (§5.2.1 / §5.2.2) — so V2.5-sigmoid is the recommended tissue prioritization axis, with V2.5-diff retained for AUC-parity use. V1 collapses on tissue (AUC 0.44 on wide, below random).

5.3.1 σ_floor sensitivity

The 0.05 floor on per-side σ is the single most influential implementation constant in the V2.5 composite (§3.1). A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} on the validated label set (reproducible via `uv run scripts/sigma_floor_sweep.py`):

cohort	σ_floor=0.02 AUC (tie@100)	σ_floor=0.05 *(default)* AUC (tie@100)	σ_floor=0.10 AUC (tie@100)	σ_floor=0.15 AUC (tie@100)
GSE322563 HM450	0.992 (719)	**0.990 (190)**	0.989 (1)	0.985 (1)
GSE322563 native v2	0.990 (1,605)	**0.986 (421)**	0.985 (1)	0.982 (1)
GSE77348	0.983 (1,019)	**0.982 (299)**	0.981 (1)	0.967 (1)
GSE69914 (tissue)	0.705 (2)	**0.773 (2)**	0.854 (1)	0.823 (4)

Matched-cell-line AUC is robust near the shipped default: movement is ≤0.02 across the 7.5× sweep, though at σ_floor = 0.15 GSE77348 drops slightly below V1 and raw Δβ-only. The stronger effect is top-K usability: on GSE322563 HM450 the K=100 tied band shrinks 719 ⟶ 190 ⟶ 1 ⟶ 1 as σ_floor increases. Tissue is genuinely σ_floor-sensitive: GSE69914 peaks at 0.854 when σ_floor = 0.10, not at the V2.5-diff default 0.05. Full per-positive grid: `examples/sigma_floor_sweep.md`.

5.3.2 δ sensitivity

δ is the only V2.5 hyperparameter selected against an empirical cohort (GSE77348, the development cohort; §4.0). A sweep over δ ∈ {0.1, 0.2, 0.3, 0.4, 0.5} on every cohort tests both matched-cell-line stability *and* the appropriateness of the development-cohort selection of δ = 0.2 on the non-development cohorts (reproducible via `uv run scripts/delta_sensitivity_sweep.py`):

cohort	δ=0.1	δ=0.2 *(default)*	δ=0.3	δ=0.4	δ=0.5
GSE322563 HM450	0.988 (tie@100=324)	**0.990 (190)**	0.991 (1)	0.985 (1)	0.985 (1)
GSE322563 native v2	0.985 (tie@100=764)	**0.986 (421)**	0.987 (36)	0.977 (1)	0.977 (1)
GSE77348 *(dev)*	0.979 (tie@100=482)	**0.982 (299)**	0.983 (5)	0.984 (1)	0.984 (1)
GSE69914 (tissue)	0.827 (tie@100=2)	**0.773 (2)**	0.707 (4)	0.668 (4)	0.643 (3)

(GSE77348 is the development cohort against which δ = 0.2 was originally selected; results at δ ≠ 0.2 on that row are *post hoc* sensitivity. The other three rows are honest sensitivity since δ was not selected against them.)

Matched-cell-line AUC is δ-stable (≤0.014 movement across the sweep). δ = 0.3 slightly improves AUC and shrinks tied bands, but δ = 0.2 is retained because it was fixed before primary evaluation and functions as conservative triage on low-`n` cohorts. Tissue shows the opposite direction for V2.5-diff: AUC drops from 0.827 at δ = 0.1 to 0.643 at δ = 0.5. This tissue-specific `p_diff` sensitivity does not transfer cleanly to gap_sigmoid (§5.2.2). Full grid: `examples/delta_sensitivity_sweep.md`.

5.4 Out-of-distribution boundary case — GSE68379

Reported here separately. ***Not*** *pooled with the primary-endpoint or sensitivity tables.*

A cross-series run against the Sanger GDSC 52-breast-line panel × GSE69914 healthy normal (n = 52 / 50) at the Roth-validated labels produced systematically inverted AUC:

label	V1	tumor_only	V2.5	V2.5 tie_band@100
validated	0.510	0.222	0.197	1,602
narrow	0.485	0.228	0.169	1,602
wide	0.407	0.462	0.269	1,602

The root cause is a probe-level biology mismatch. At the three Roth Fig. 5d target probes:

probe	gene	Roth MCF-7 β (GSE322563)	Sanger MCF-7 β (GSE68379)
cg05251676	EGFLAM	0.01	**0.92**
cg25338972	ESR1	0.07	**0.63**
cg01364137	GATA3	0.02	**0.70**

Same cell line name, opposite methylation state at the validated loci. MCF-7 has accumulated substantial genetic and epigenetic drift across laboratories over decades of separate passaging. Because the Roth labels encode "tumor hypomethylated at target" and Sanger's MCF-7 is methylated at those targets, any scorer whose composite contains `p_targ = P(β_tumor < 0.30)` will rank the validated positives at the *bottom* of its distribution — which is what V2.5 and `tumor_only` do. The inversion is the correct response of the scorer to the underlying biology; the labels are the component that is no longer valid on this cohort.

GSE68379 is reported as a label-transportability boundary, not a failed generalization of V2.5. Pooling it into an average across cohorts would misrepresent the result; redefining positives from GSE68379′s own β values would be circular (labels derived from the data under test). It remains in the documentation as a documented boundary case.

5.5 Top-hit annotation (tie-window-aware)

An annotation pass (nearest gene, TSS distance, feature class, CpG island context) was run on each benchmark-selected top-20 window. On low-`n` cell-line cohorts, those "top-20" lists are not stable ranked shortlists: they are 20-record windows inside much larger tied bands selected by the documented `candidate_id` tie-break. The four genes shared between the GSE322563 and GSE77348 V2.5-diff top-20 windows (ADRB1, CELF2, GFRA1, XPNPEP1) should therefore be read as window convergence inside tied bands, not convergence of the underlying ranking. On GSE69914 tissue (`tie_band@100 = 2`), top-20 membership is much closer to a genuine ordering and maps to a disjoint tissue-specific set (including RUNX2, SCML4, FOXC1/FOXCUT, MAS1L, COL21A1, MXI1, TENM2, MAT2B). Full annotated TSVs are committed under `examples/*/top20_annotated_*.tsv`. None of the three Roth-validated probes appears in any top-20; the primary endpoint is rank lift/AUC, not top-20 recovery.

5.6 p_trust sensitivity — base weights and ramp_n

Validated-label AUC is exactly invariant to `ramp_n ∈ {10, 20, 30, 50, 100}` within each cohort. The reason is structural: the `p_trust` ramp is `min(1, min(n_t, n_n) / ramp_n)`; with per-cohort uniform `(n_t, n_n)` (verified across 100k sampled records on every cohort: a single distinct pair), `ramp_n` reduces to a global multiplicative scale on every record's `p_therapeutic_selectivity`, which is rank-preserving. AUC is also near-invariant ($\Delta \leq 0.001$) to the per-`EvidenceClass` base-weight set on the three matched cell-line cohorts, because their top-100 windows are 100/100 EXACT records (§5.7 below). On GSE69914 tissue (top-100 =

33 EXACT / 67 PROXIMAL_CLOSE), AUC moves up to ±0.006 across an aggressive (0.95/0.85/0.65/0.35), shipped (0.95/0.75/0.45/0.15), and conservative (0.99/0.50/0.20/0.05) base-weight set — small enough that the shipped setting is within `≲ 0.006` of the per-cohort best on every cohort × tier combination tested. Full sweep in `examples/p_trust_sensitivity_sweep.{tsv,md}` at this tag; reproduce via `uv run python scripts/p_trust_sensitivity_sweep.py`.

5.7 EvidenceClass distribution — full catalogs vs top-100 windows

The `p_trust` factor's discrete `EvidenceClass` lever (§3.5) controls how strongly a candidate's evidence weight is scaled by distance from the nearest assayed CpG. To make that lever's behavior auditable, we report the per-class composition of each cohort's full chr5/6/10 catalog and of the top-100 window selected by the V2.5-diff axis (committed in `examples/evidence_class_distribution.{tsv,md}` at this tag; reproduced via `uv run python scripts/evidence_class_distribution.py` after the per-cohort scored JSONL chain in the reproducibility appendix).

Full-catalog distributions are dominated by `regional` records (~56% on the HM450 cohorts, ~74% on GSE322563 native EPIC v2 and on GSE69914 tissue) with `exact` records the smallest class (~1.1%–1.4% of the catalog). The top-100 windows are strongly enriched: 100% of the matched-cell-line top-100 are `exact`-class records (a ~70× enrichment over the catalog baseline), and the GSE69914 tissue top-100 splits 33 EXACT / 67 PROXIMAL_CLOSE — broader because the tissue cohort's higher `n` lifts records away from `p_trust` saturation (per-class median `p_trust` 0.95 EXACT vs 0.75 PROXIMAL_CLOSE on tissue, vs 0.06–0.10 EXACT on the matched cell-line cohorts where `n_t = n_n = 2/2 or 3/3` puts the saturating `min(1, n / ramp_n)` factor near zero). V2.5-sigmoid's top-100 EvidenceClass mix is structurally identical because both V2.5 variants share the same outer `p_targ × ... × p_trust` factor; only the gap-factor slot differs.

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6 · Discussion

6.1 Recommended use

Use case	Recommended axis	Rationale
New probabilistic discovery runs on tested non-boundary regimes	**V2.5-sigmoid** (`tumor_plus_gap_sigmoid`)	AUC-equivalent to V2.5-diff on matched cell lines, higher AUC on tissue, and `tie_band@100 = 1` on every matched-cell-line WG cohort (§5.2.2).
Backward-compatible probabilistic runs	V2.5-diff (`tumor_plus_differential_protection`)	Retained for AUC parity with earlier scored JSONLs; not preferred for new WG-scale discovery because low-`n` tied bands expand to 421–1,493 records.
Stable package default / deterministic top-K	V1 `final_score`	Continuous score with `tie_band = 1`; retained for backward compatibility and deterministic lists, not AUC leadership.
Diagnostic ablation	V2 `tumor_only`	Useful AUC sanity check; not a discovery axis because top-K collapses into thousands-record tied bands.
Cross-series label transport (GSE68379-like)	Unsupported	Cell-line drift can invert label logic; do not pool with discovery-mode benchmarks.

On low-`n` matched cell-line cohorts, the V2.5 generation should be read as a **rank-lift tool**, not a stable top-K shortlist generator. It lifts the three Roth positives into the top percentile of millions of candidates (§5.1), but the visible top-20 can be a window inside a hundreds-record tied band. Selecting individual wet-lab candidates from that class requires secondary evidence such as annotation, chromatin context, guide quality, and off-target risk.

6.2 Limitations

1. **Small validated set.** The primary endpoint has three positives. AUC is therefore rank lift on three sites, not generalized discovery evidence; per-positive ranks are reported in §5.1.
2. **Low-replicate tied bands.** For `n < ramp_n = 30`, probabilistic scores can form hundreds-to-thousands-record tied bands. The benchmark reports `tie_band_size_at_k` and adversarial P@K intervals so top-K claims remain explicit.
3. **Catalog and platform scope.** §5.1 uses chr5/6/10 HM450-scope negatives because the validated Roth targets live there; §5.2.2 extends the denominator to frozen WG HM450 and native EPIC v2 catalogs. Random-chromosome-matched controls remain future work.
4. **Cell-line drift.** GSE68379 shows that same-named MCF-7 stocks can have opposite methylation at Roth targets. Cohort-specific methylation must be verified before editing follow-up.

5. **V2.5-diff σ_floor.** The 0.05 per-side σ floor is a V2.5-diff constant. It is robust on matched cell lines but tissue-sensitive (§5.3.1). The current recommendation is V2.5-sigmoid, whose fixed-bandwidth behavior is evaluated separately in §5.2.2.
6. **Top-hit annotation is descriptive.** Low-`n` top-20 lists are tie-window slices, not stable rankings. They require secondary biological filters before wet-lab selection.
7. **No prospective wet-lab validation.** The claim is scoring and tie-band-aware benchmarking on public methylation data, not editing efficiency at named sites.

6.3 Next steps

1. Prospective wet-lab editing validation of top V2.5-sigmoid candidates.
2. Random-chromosome-matched negative-universe controls to complement the completed WG panel.
3. A one-shot `regime` YAML selector on top of the shipped `tumor_plus_gap_sigmoid` mode.
4. Alternative gap factors: SE-on-mean `p_diff`, exact difference-of-Betas, or mixture-aware variants.
5. Continuous observation-confidence modeling beyond discrete `EvidenceClass` bins.
6. A second independent-lab MCF-7 / MCF-10A EPIC cohort if one becomes public.
7. Formal SCREEN cCRE integration and an R `limma` parity check.

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7 · Methods — key equations

7.0 Canonical axis-naming table

With two V2.5-generation modes and four baselines, the paper uses the following canonical names; enum names are what `CohortConfig` accepts in YAML, prose names are what §5 tables and recommendations use:

Prose name	Code enum / field	Role	Recommendation
V1	`final_score`	stable-release default; `tie_band = 1` by construction	backward compatibility + deterministic top-K
V2	`probabilistic_mode: tumor_only`	first-pass composite; `p_prot` anti-predictive on tissue	diagnostic / audit only
V2.5-diff	`probabilistic_mode: tumor_plus_differential`	diagnostic V2.5 `p_targ × p_diff × p_trust`	retained for backward compat + AUC parity on cell-line cohorts
V2.5-sigmoid	`probabilistic_mode: tumor_plus_gap_sigmoid`	recommended V2.5 variant; `p_targ × p_gap_sigmoid × p_trust`	**recommended probabilistic prioritization axis** (all tested non-boundary regimes)
Δβ-only	`naive_selectivity` (score-field)	raw methylation gap	honest baseline
Δβ_z	computed post-hoc from scored JSONL	σ-normalized raw gap	uncertainty-aware ablation baseline
limma-style moderated-t	probe-level Smyth 2004 eBayes	DMR comparator	methods-journal-standard baseline

The terms "probability-scale selectivity score" (paper-level) and the field `p_therapeutic_selectivity` (code-level) refer to the same emitted scalar under every probabilistic mode. "V2.5" without a `-diff` / `-sigmoid` qualifier refers to the *generation* (`tumor_plus_*`) and is used when a statement applies to both variants.

Key probabilistic factors:

```
p_targ = P(β_tumor < 0.30)   via method-of-moments Beta(α, β) fit,
p_prot = P(β_normal ≥ 0.50)  via same CDF dispatch.     [V2 only]
p_diff = 1 − Φ((δ − (μ_n − μ_t)) / σ_Δ)
         where σ_Δ = √(max(σ_t, 0.05)² + max(σ_n, 0.05)²)
                           and σ_k = IQR_k / 1.349.      [V2.5-diff]
p_gap_sigmoid = 1 / (1 + exp(−((μ_n − μ_t) − δ) / σ_fixed))
         (δ = 0.2, σ_fixed = √2 × σ_floor ≈ 0.0707 by default;
          σ_fixed strictly positive, enforced at CohortConfig
          boundary). Numerically-stable closed form, pure stdlib.
                                                      [V2.5-sigmoid]
p_trust = EvidenceClass.base × min(1, min(n_t, n_n) / 30)
         (EvidenceClass.base ∈ {0.95, 0.75, 0.45, 0.15, 0.0}).
```

V1 uses a deterministic sequence × selectivity × confidence score minus heterogeneity and low-coverage penalties; it is continuous-valued and therefore has `tie_band = 1` in every benchmark here. Probabilistic composites by mode:

```
tumor_only                          : p_targ × p_trust
tumor_plus_normal_protection        : p_targ × p_prot × p_trust
tumor_plus_differential_protection  : p_targ × p_diff × p_trust
    (= V2.5-diff)
tumor_plus_gap_sigmoid              : p_targ × p_gap_sigmoid × p_trust
    (= V2.5-sigmoid; recommended probabilistic prioritization axis, §5.2.2)
```

Validators enforce mode-specific audit fields and exact composite consistency for the emitted `p_therapeutic_selectivity` scalar.

Benchmark ranking contract. Let the scored set be N records, each with a score `s_i` and a ground-truth label `y_i ∈ {0, 1}` (1 for positive). The benchmark sorts records by the total order

```
    (s_i, c_i)   with primary  -s_i  ascending
                       secondary c_i   ascending
```

where `c_i` is the candidate_id. Let `r_1, r_2, …, r_N` be the resulting ranking (smallest `(-s, c)` first). Define:

```
    top-K               := {r_1, …, r_K}
    cutoff_score        := s at position K in the sort, i.e. s_{r_K}
    tied_band           := {r_i : s_{r_i} = cutoff_score}           # records at the
cutoff
    tie_band_size_at_k  := |tied_band|
    k_band              := |tied_band ∩ top-K|                      # records in top-K at
the cutoff
    drop_slots          := tie_band_size_at_k − k_band              # records at cutoff
outside top-K
    band_pos            := |{r_i ∈ tied_band : y_{r_i} = 1}|
    committed_pos       := |{r_i ∈ top-K \ tied_band : y_{r_i} = 1}|
```

Then:

```
P@K_observed := (committed_pos + |{r_i ∈ tied_band ∩ top-K : y_{r_i} = 1}|) / K

P@K_min      := (committed_pos + max(0, band_pos − drop_slots)) / K
P@K_max      := (committed_pos + min(band_pos, k_band))          / K
```

`P@K_min` pushes as many tied-band positives as possible outside top-K; `P@K_max` pulls as many as possible inside top-K. By construction:

```
P@K_min ≤ P@K_observed ≤ P@K_max
```

and the interval collapses when `tie_band_size_at_k = 1`. Recall uses `n_positives` as the denominator. AUC is Mann-Whitney U / (`n_pos · n_neg`) with ties contributing 0.5.

Data and code availability

- **Code**: https://github.com/AllisonH12/thermocas9. Cite tag **`memo-2026-04-22-av`** for this document. 245 tests pass under `uv run pytest -q`.
- **Citable archive (DOI)**: a Zenodo release archive of the tagged revision is planned at the time of preprint posting; the GitHub → Zenodo integration mints a DOI for each GitHub release tag. The DOI will be added to this section and to the citation block below before journal-version submission. Until then, the immutable git tag above is the canonical citable identifier.
- **Cohort data**: publicly-downloadable GEO series GSE322563, GSE77348, GSE69914, GSE68379; build scripts in `scripts/build_gse*_cohort.py` produce the per-probe summary TSVs in `data/derived/*_cohort/`. Positives-list builder at `scripts/build_roth_positives.py` (requires the Ensembl REST `/map` endpoint for the hg38 → hg19 liftover of Roth Fig. 5d coordinates).
- **Reference data**: UCSC hg19 `refGene.txt.gz` and `cpgIslandExt.txt.gz` (fetched on demand; gitignored).
- **Benchmark artifacts**: every `BenchmarkResult` JSONL row under `examples/*_roth_labels/` carries `precision_at_k`, `precision_at_k_{min,max}`, `recall_at_k`, `recall_at_k_{min,max}`, `roc_auc`, `tie_band_size_at_k`, and `tie_break_policy`.
- **Annotated top-20 TSVs**: under `examples/*/top20_annotated_*.tsv`.

Reproducing the cross-cohort matrix

The exact build, score, benchmark, and aggregation commands are kept in the scripts referenced throughout §4–§5 and in the committed provenance files under `data/derived/*PROVENANCE*`. Large scored JSONLs are gitignored but reproducible from the public GEO inputs, frozen catalogs, cohort YAMLs, and scripts in this tag.

Citation

If this paper or the associated code is useful in your work, please cite the upstream paper:

> Roth M.O., Shu Y., Zhao Y., Trasanidou D., Hoffman R.D., et al. Molecular basis for methylation-sensitive editing by Cas9. *Nature* (2026). DOI 10.1038/s41586-026-10384-z.

For the probabilistic scoring framework specifically, cite this paper plus the code URL above.

Acknowledgements

This work describes a framework built as an educational / research project. ThermoCas9′s biochemical characterization and the validated target sites are entirely due to Roth et al.; any errors in the re-implementation of the scoring or benchmarking are the author's.